TIC 412843260

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

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BLS EPHEMERIS
  Period       2.090432 d  (ALIASED, see below)
    -> corrected period: 4.180865 d
  Epoch (t0)   1491.8718 BTJD
  Duration     0.1700 d
  Depth        2.174 %  (21,738 ppm)
  SNR          177.8

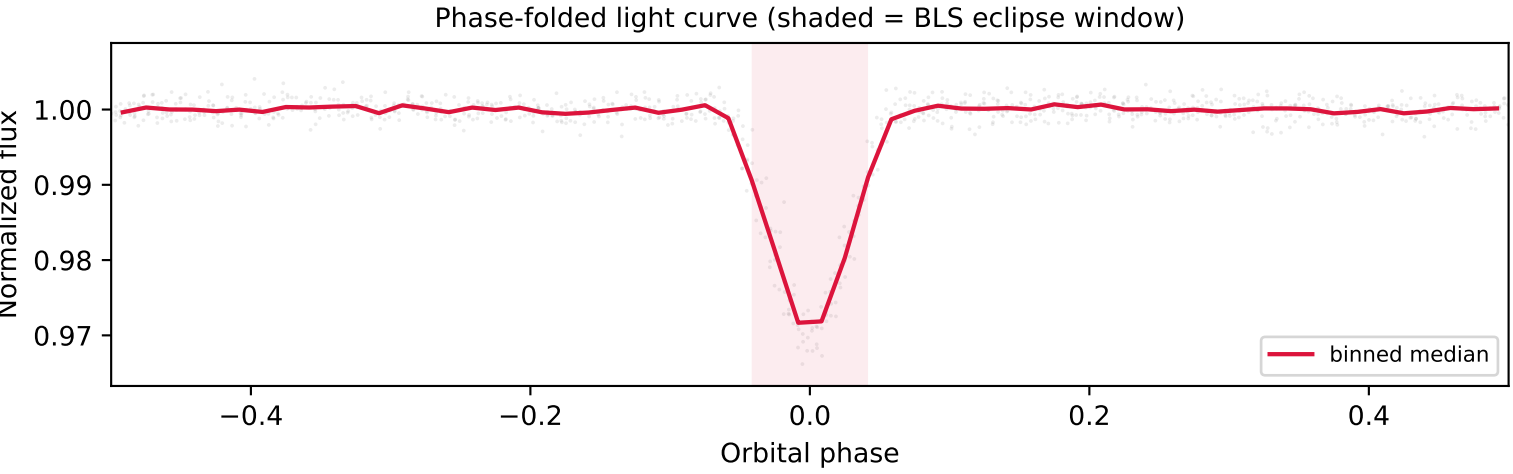
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     11  (gate >= 3)
  Robust depth ratio    1.06  (gate >= 0.50)
  Robust depth sigma    142.4  (gate >= 7.0)
  Odd/even z-score      4.33  (ALIAS FLAGGED)
  Secondary eclipse     not detected
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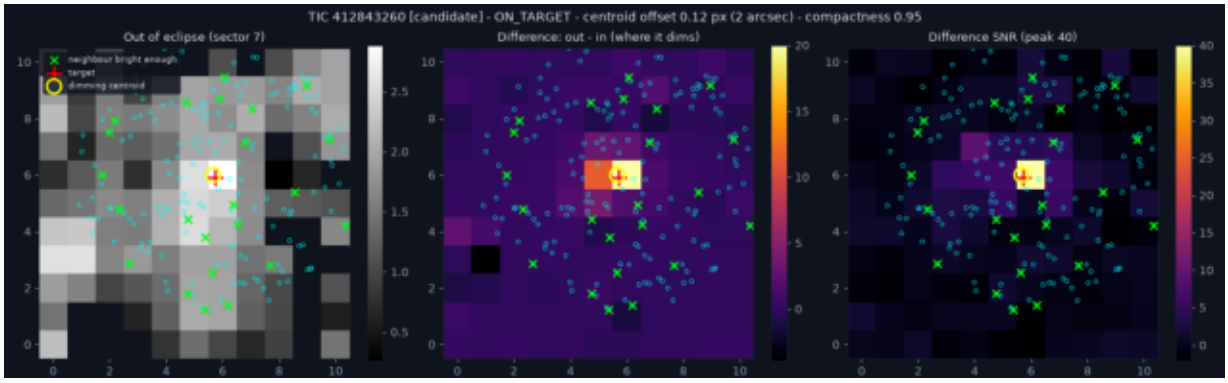
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 143.9 sigma    t0 refined by -0.0075 phase
  Difference-image peak SNR   40.04   (gate >= 5.0)      Compactness   0.95   (gate >= 0.50)
  Dimming centroid offset     0.12 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 412843274
    1.29 px away, dTmag +3.89 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.33 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 4.180865 d, in any write-up.